

UPSC Smart MCQ Builder & Practice Portal

Project Overview

Project Name

UPSC Smart MCQ Builder & Practice Portal

Project Type

Personal Web Application for UPSC Prelims Preparation

Purpose

The purpose of this application is to provide a centralized platform for creating, managing, practicing, and analyzing UPSC multiple-choice questions. The platform is designed for a single user and focuses on minimizing the effort required to build large question banks through automatic option generation.

The system will allow creation of unlimited sections, topics, and questions while providing quiz functionality, performance tracking, revision tools, bookmarks, and analytics.

Project Goals

Primary Goals

- Create unlimited UPSC MCQs
- Organize questions by subject and micro-topic
- Automatically generate incorrect options
- Conduct section-wise and topic-wise tests
- Calculate scores instantly
- Review mistakes
- Track performance trends
- Support revision and bookmarking

Secondary Goals

- Reduce manual data entry
 - Improve revision efficiency
 - Maintain a personal UPSC knowledge database
 - Enable quick search and filtering
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User Type

Admin User

Only one user will use the platform.

Capabilities:

- Login
- Create Sections
- Create Topics
- Create Questions
- Edit Questions
- Delete Questions
- Attempt Quizzes
- Review Performance
- Export Data
- Backup Database

No registration system is required.

Website Structure

Dashboard → Section Management → Topic Management → Question Builder → Question Bank → Quiz Module → Results → Review Answers → Analytics → Bookmarks → Revision Mode → Settings

Login Page

Purpose

Protect access to personal study data.

Components

- Username
- Password
- Login Button

Features

- Admin-only login
- Remember session

- Logout option
-

Dashboard

Purpose

Display overall preparation status.

Statistics

- Total Questions
- Total Sections
- Total Topics
- Questions Attempted
- Accuracy Percentage
- Bookmarked Questions
- Recent Quiz Activity

Quick Actions

- Add Question
 - Start Quiz
 - Review Mistakes
 - View Analytics
-

Section Management

Purpose

Organize major UPSC subjects.

Default Sections

- Environment
- Modern History
- Ancient History
- Medieval History
- Polity
- Geography
- Economy
- Science & Technology
- International Relations
- Art & Culture

- Current Affairs
- Ethics
- Miscellaneous

Features

- Add Section
 - Rename Section
 - Delete Section
 - View Question Count
 - View Topic Count
-

Topic Management

Purpose

Organize micro-topics inside sections.

Example

Environment

- National Parks
- Climate Change
- Wetlands
- Biodiversity
- Environmental Conventions
- Protected Areas

Polity

- Parliament
- Judiciary
- Fundamental Rights
- DPSP
- Constitutional Amendments

Features

- Add Topic
 - Rename Topic
 - Delete Topic
 - Unlimited Topics
-

Question Builder

Purpose

Create questions efficiently.

Manual Question Entry

Fields

- Question
- Option A
- Option B
- Option C
- Option D
- Correct Answer
- Difficulty
- Tags

Example:

Question: Kaziranga National Park is located in which state?

Options: Assam Odisha Tripura Kerala

Correct Answer: Assam

Smart Question Entry

Fields

- Question
- Correct Answer

System automatically generates three incorrect options from previously stored answers.

Example:

Question: Kaziranga National Park is located in which state?

Correct Answer: Assam

Generated Options:

- A. Assam
 - B. Kerala
 - C. Rajasthan
 - D. Odisha
-

Smart Option Generator

Purpose

Reduce MCQ creation effort.

Working

Every unique answer entered by the user is stored in an Answer Pool.

Example:

Assam Kerala Odisha Tamil Nadu UNEP UNESCO Lok Sabha Rajya Sabha

When a new question is added:

Correct Answer: Assam

The system randomly selects three additional answers from the Answer Pool.

Rules

- No duplicate options
 - One correct answer only
 - Random option order
 - Automatic shuffling
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Question Bank

Purpose

Store and manage all questions.

Features

- Search Questions
 - Edit Questions
 - Delete Questions
 - Duplicate Questions
 - Filter by Section
 - Filter by Topic
 - Filter by Difficulty
 - Filter by Tags
-

Bulk Question Upload

Purpose

Add many questions quickly.

Input Format

Question: Kaziranga National Park is located in which state?

Answer: Assam

Question: Project Tiger was launched in which year?

Answer: 1973

Features

- Bulk Save
 - Automatic Option Generation
 - Validation Checks
-

Quiz Module

Purpose

Practice questions in exam mode.

Quiz Setup

Select:

- Section
- Topic
- Difficulty
- Number of Questions

Examples:

Environment → 50 Questions

Polity → 100 Questions

Mixed Test → 100 Questions

Quiz Interface

Components

- Question Number
- Question Text
- Four Options
- Previous Button
- Next Button
- Submit Button

Optional Features

- Timer
 - Question Palette
 - Flag for Review
-

Results Page

Purpose

Display performance after quiz completion.

Metrics

- Total Questions

- Correct Answers
- Wrong Answers
- Unattempted Questions
- Score Percentage

Example:

Total Questions: 50

Correct: 42

Wrong: 8

Score: 84%

Review Answers Page

Purpose

Analyze mistakes.

Displays

- Question
- Selected Answer
- Correct Answer
- Status

Color Coding

Green: Correct

Red: Incorrect

Analytics Dashboard

Purpose

Track learning progress.

Overall Analytics

- Questions Attempted
- Accuracy Rate
- Improvement Trend
- Total Study Time

Section Analytics

Environment: 88%

Polity: 76%

History: 69%

Topic Analytics

Wetlands: 91%

Climate Change: 71%

National Parks: 84%

Search Module

Purpose

Quickly locate questions.

Search By

- Keyword
- Section
- Topic
- Correct Answer
- Tags

Example:

Search: Ramsar

Returns all Ramsar-related questions.

Bookmark System

Purpose

Save important questions.

Features

- Bookmark Question
 - Remove Bookmark
 - Practice Bookmarked Questions
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Revision Mode

Purpose

Improve retention.

Revision Categories

- Wrong Questions
 - Bookmarked Questions
 - Weak Topics
 - Recently Added Questions
 - Random Revision
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Settings Page

Features

- Change Password
 - Export Database
 - Import Questions
 - Backup Database
 - Restore Database
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Database Architecture (SQLite)

Database File:

Sections Table

Fields:

- id
 - section_name
 - created_at
 - updated_at
-

Topics Table

Fields:

- id
 - section_id
 - topic_name
 - created_at
 - updated_at
-

Questions Table

Fields:

- id
 - topic_id
 - question_text
 - correct_answer
 - difficulty_level
 - tags
 - created_at
 - updated_at
-

Options Table

Fields:

- id
- question_id
- option_text

- is_correct
-

Answer Pool Table

Purpose: Store unique answers for automatic option generation.

Fields:

- id
 - answer_text
 - category
 - created_at
-

Quiz Results Table

Fields:

- id
 - score_percentage
 - total_questions
 - correct_answers
 - wrong_answers
 - completed_at
-

User Answers Table

Fields:

- id
 - quiz_result_id
 - question_id
 - selected_option
 - correct_option
 - is_correct
-

Bookmarks Table

Fields:

- id
- question_id

- bookmarked_at
-

Revision History Table

Fields:

- id
 - revision_type
 - questions_attempted
 - score_percentage
 - completed_at
-

Backup System

Manual Backup

Export:

upsc.db

Automatic Backup

Weekly backups:

backup_YYYY_MM_DD.db

Export Formats

- CSV
 - Excel
 - JSON
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Technology Stack

Frontend

- React

- TypeScript
- Tailwind CSS

Backend

- Node.js
- Express.js

Database

- SQLite

Authentication

- Admin Login

Hosting

- Railway
 - Render
 - VPS
-

Future Enhancements

AI Distractor Generator

Generate UPSC-quality wrong options automatically.

Example:

Correct: Kaziranga

Generated:

- Manas
 - Nameri
 - Dibru-Saikhowa
-

Previous Year Question Mode

Tags:

- UPSC 2023

- UPSC 2024
 - UPSC 2025
 - CAPF
 - State PCS
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Spaced Repetition System

Automatically reschedule weak questions.

Difficulty Rating

- Easy
 - Medium
 - Hard
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Mobile App Version

Future React Native application using the same SQLite-backed API.

Expected Outcome

The final product should function as a complete personal UPSC preparation platform that allows creation of unlimited sections, topics, and questions, supports automatic MCQ generation, conducts quizzes, tracks performance, provides revision tools, and stores all data securely using SQLite while remaining simple, fast, and inexpensive to host.